Customer Segmentation Using RFM, K-Means, and DBSCAN

Online Retail dataset | Cleaning, RFM features, model validation, segments and recommendation

1. Data Cleaning Summary

The raw dataset has **541,909** transactions and 8 columns. The **5,268** duplicate rows were removed first (`df.drop_duplicates()`), leaving 536,641 rows. The issues below were counted on the de-duplicated data, and the cleaning rules were then applied in the notebook (Sections 2-3).

Issue found in data	Count	Action taken
Duplicate rows	5,268	Rows removed (df.drop_duplicates(), notebook Section 2.3)
Missing CustomerID	135,037	Rows removed (customer-level analysis needs an ID)
Missing Description	1,454	Not needed for RFM; none remain after the CustomerID filter
Cancelled invoices (InvoiceNo starts with "C")	9,251	Rows removed
Negative Quantity	10,587	Rows with Quantity ≤0 removed
Zero / negative UnitPrice	2,512	Rows with UnitPrice ≤0 removed

After cleaning	Value
Retained transactions	392,692
Retained invoices	18,532
Retained customers	4,338
Reference date (latest invoice)	2011-12-09 12:50

2. RFM Definition

Feature	Definition (as computed in the notebook)
Recency	Days between the customer's latest invoice date and the dataset's latest invoice date (lower = more recent)
Frequency	Number of distinct non-cancelled invoices per customer
Monetary	Sum of Quantity x UnitPrice across the customer's transactions

Median customer: Recency 50 days, Frequency 2 invoices, Monetary 668.57. Maximums are far higher (Frequency 209, Monetary 280,206), a sign of strong right skew.

CustomerID	Recency	Frequency	Monetary
12346	325	1	77,183.60
12347	1	7	4,310.00
12348	74	4	1,797.24
12349	18	1	1,757.55
12350	309	1	334.40

Table 1. First five rows of the RFM table (4,338 customers).

3. Distribution, Transformation and Scaling

Histograms of the raw RFM features show that Frequency and Monetary are concentrated near zero with a very long right tail, while Recency is right-skewed but milder.

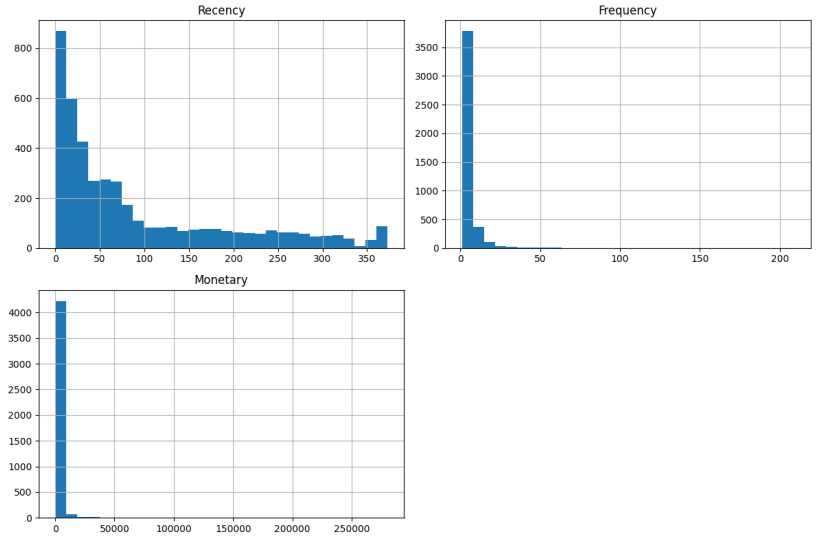


Figure 1. Histograms of Recency, Frequency and Monetary (output of the notebook, Section 5.1).

Feature	Skewness before	Skewness after (log1p)
Recency	1.246	not transformed
Frequency	12.067	1.209
Monetary	19.339	0.397

log1p was applied to Frequency and Monetary. It cut Monetary skewness from 19.3 to 0.40 and Frequency from 12.1 to 1.21 (a smaller improvement). The three clustering features - **Recency, Frequency_log, Monetary_log** - were then standardized with StandardScaler (CustomerID excluded) so that no feature dominates the distance calculations of K-Means and DBSCAN. The scaled matrix has shape (4,338, 3).

4. K-Means Validation

K-Means was run for k = 2 to 8 (random_state = 42, n_init = 10). Inertia always falls as k grows; the Silhouette Score peaks at **k = 3 (0.416)**.

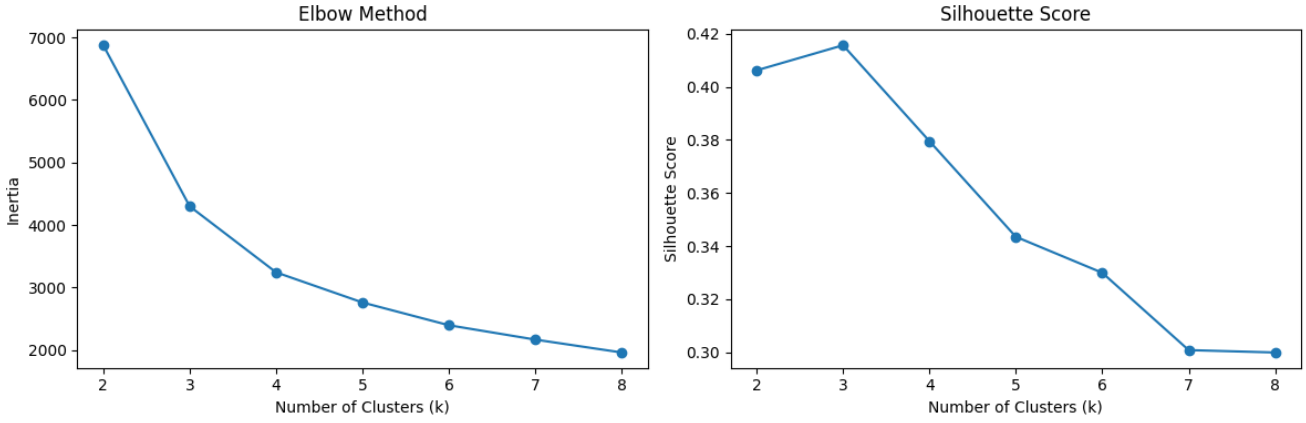


Figure 2. Elbow (inertia) and Silhouette Score versus k (output of the notebook, Section 6.3).

k	2	3	4	5	6	7	8
Inertia	6874.9	4299.1	3243.0	2760.8	2396.9	2167.0	1961.1
Silhouette	0.406	**0.416**	0.380	0.343	0.330	0.301	0.300

Table 2. K-Means diagnostics. k = 3 was chosen as the best Silhouette among the tested values with a manageable, interpretable number of segments; it is not claimed to be the only valid solution.

5. DBSCAN Comparison

DBSCAN was fitted on the same three scaled features with three parameter settings.

eps	min_samples	Clusters	Noise	Silhouette
0.5	5	2	1.1%	0.3268
1.0	5	1	0.2%	not applicable
1.5	10	1	0.1%	not applicable

Table 3. DBSCAN sensitivity. Silhouette needs at least 2 clusters; the eps = 0.5 score excludes noise points, so it is not directly comparable to the K-Means score.

Larger eps values made points density-connected into one large cluster, so the result is very sensitive to parameters. K-Means assigns every customer to a segment and gave three distinguishable RFM profiles (Silhouette 0.416); DBSCAN gave a lower score (0.327) with two clusters at best and one cluster otherwise.

6. K-Means Segment Profile (k = 3)

Cluster	Size	Median Recency (days)	Median Frequency	Median Monetary	Segment name
0	1,320	16	7	2,570.50	Recent, frequent, higher-spend
1	981	252	1	290.51	Less-recent, infrequent, lower-spend
2	2,037	44	2	508.80	Regular / lower-spend

Table 4. Segment sizes and median RFM values (sizes sum to 4,338).

7. PCA Visualization

PCA projects the three scaled features onto two components for display only; clustering itself was done on the three scaled features. Some overlap between segments is expected because two dimensions do not preserve all information.



Figure 3. K-Means segments in PCA space (output of the notebook, Section 9).

8. Recommendation

Use K-Means with k = 3. It has the highest Silhouette Score among the tested K-Means models (0.416), covers every customer, and yields three segments that map to measurable engagement and retention actions. DBSCAN is useful for density-based exploration and noise detection, but none of the tested settings produced equally useful segments here.

Segment	Proposed action (to be tested)	How to measure
Cluster 0 - Recent, frequent, higher-spend (1,320)	Test a loyalty / rewards program	Retention rate versus reward cost
Cluster 2 - Regular, lower-spend (2,037)	Test targeted offers that encourage repeat purchases	Incremental purchase activity versus a control group
Cluster 1 - Less-recent, infrequent, lower-spend (981)	Run a low-cost reactivation campaign	Return-to-purchase rate versus a holdout group

9. Limitations

- Segments describe observed transaction behavior only; they do not imply personal traits, motivations or future behavior.
- The proposed actions are recommendations to be tested, not guaranteed outcomes.
- Silhouette values around 0.4 indicate moderate separation, and the segments overlap in the PCA view.
- The DBSCAN silhouette excludes noise points while the K-Means silhouette includes all customers, so the two are not directly comparable.
- Results depend on the chosen features (Recency, log Frequency, log Monetary), the log1p transformation and the DBSCAN parameters tested.
- Cluster numbers come from a K-Means run and can change between runs or datasets; segments should be identified by their RFM profile, not by cluster number.